

Appendix B

Revision History

This appendix provides a list of the major differences between revisions of the *e200z760n3 Power Architecture® Core Reference Manual*.

B.1 Changes between revisions 1 and 2

Table B-1. Changes between revisions 1 and 2

Chapter	Description
Chapter 3, "Instruction Model"	In Section 3.14, "Volatile Context Save/Restore Unit" For instruction "e_lmvsprw" added NOTE: "If the EA is misaligned and the e_lmvsprw is followed by either a branch to link register or branch to count register within 4 instructions, the core can lock up during exception handling for the misalignment. To avoid this issue, do not do misaligned on e_lmvsprw or ensure there are at least 4 instructions in between the e_lmvsprw and the branch to LR or CTR. This issue does not apply to Book E applications."
Chapter 4, "Instruction Pipeline and Execution Timing"	Section 4.3.2, "Instruction Prefetch Buffers and Branch Target Buffer" - Added NOTE: - The e200z7 core can prefetch up to 2 cache lines (64 bytes total) beyond the current instruction execution point. Executing code within the last 64 bytes of a memory region such as internal SRAM or Flash may cause a bus error when pre-fetching occurs past the end of memory. Do not place code to be executed within the last 64 bytes of a memory region. - An ECC exception can occur if pre-fetches occur at locations that are valid but not yet initialized for ECC. When executing code from internal ECC SRAM, initialize memory beyond the end of the code until the next 32-byte aligned address and then an additional 64 bytes to ensure that pre-fetches cannot land in uninitialized SRAM. - The Boot Assist Module (BAM) is located at the end of the address space and so may cause instruction pre-fetches to wrap-around to address 0 in internal flash memory. If this first block of flash memory contains ECC errors, such as from an aborted program or erase operation, a machine-check exception will be asserted. At this point in the boot procedure, exceptions are disabled, but the machine-check will remain pending and the exception vector will be taken if user application code subsequently enables the machine check interrupt. To guard against the possibility of the BAM causing a machine-check exception to be taken, user application code should write all 1s to the Machine Check Syndrome Register (MCSR) to clear it before enabling the machine check interrupt." - Added NOTE: "Under certain conditions, if a static branch prediction and a dynamic return prediction (which uses the subroutine return address stack) occur simultaneously in the BTB, the e200z7 core can issue an errant fetch address to the memory system (instruction fetched from wrong address). This can only happen when the static branch prediction is "taken" but the branch actually resolves to "not taken". To prevent the issue from occurring, set BUCSR[BPRED] = 0b11 to configure static branch prediction to "not taken". This issue does not apply to VLE."
Section Appendix B, "Revision History"	Added Section B.1, "Changes between revisions 1 and 2"

B.2 Changes between revisions 0 and 1

Table B-2. Changes between revisions 0 and 1

Chapter	Description
Overall	Editorial changes
Chapter 1, “e200z7 Core Complex Overview”	Section 1.1, “e200z7 Overview” - Deleted paragraph “The e200z7 platform can be configured in specific processor implementations...” - In Figure 1-1, “e200z7 Block Diagram”, corrected “Single-Instruction, In-Order Dispatch” to “Dual-Instruction, In-Order Dispatch” and “Single-Instruction, In-Order Write Back” to “Dual-Instruction, In-Order Write Back” Section 1.1.1, “Features” - Changed “Branch <u>acceleration</u> using a branch target buffer (BTB)” to “Branch <u>target prefetching</u> using a branch target buffer (BTB)”. - Changed “<u>Two</u>-cycle load latency” to “<u>Three</u>-cycle load latency”. - Changed “Big- and little-endian support <u>on a per-page basis</u>” to “Big- and little-endian support”. Section 1.2.1, “Register Set” - In Figure 1-2, “e200z760 Supervisor Mode Programmer’s Model”, corrected misaligned register names for Memory Management Registers, Control & Configuration. Section 1.2.2, “Instruction Set” - In Table 1-2, “Scalar and Vector Embedded Floating-Point Instructions”, added missing floating-point instructions (efscfh/evfscfh, efscth/evfscth, efsmax/evfsmax, efsmin/evfsmin, efssqrt/evfssqrt, evfsaddsub, evfsaddsubx, evfsaddx, evfsdiffsum, evfsdiff, evfsmule, evfsmulo, evfsmulx, evfssubaddx, evfssubx, evfssubadd, evfssumdiff, and evfssum). Section 1.2.3.3, “Interrupt Types” - In Table 1-3, “Interrupt Types”, - For the Description of the Critical interrupts Category, changed “If the <u>debug interrupt</u> is not enabled, <u>it is also</u> treated as a critical interrupt.” to “If the <u>debug feature</u> is not enabled, <u>a debug interrupt</u> is treated as a critical interrupt.” - For the Programming Resources of the Debug interrupts Category, changed “Can be masked by the <u>machine check</u> enable bit, MSR[DE]. Includes the debug <u>syndrome</u> register (DBSR).” to “Can be masked by the <u>debug interrupt</u> enable bit, MSR[DE]. Includes the debug <u>status</u> register (DBSR).” Section 1.2.3.4, “Interrupt Registers” - In Table 1-4, “Interrupt Registers”, added entry for the DBSR (Debug status register). Section 1.3, “Microarchitecture Summary” - Changed “Prefetched instructions are placed into an instruction buffer <u>capable of holding six instructions</u>.” to “Prefetched instructions are placed into an instruction buffer.” - Deleted mistakenly repeated paragraph that starts with “Conditional branches which are not taken...” Section 1.3.2, “Integer Unit Features” - Changed “Divider logic for signed and unsigned divide in 4 to 15 clocks with minimized execution timing” to “Divider logic for signed and unsigned divide in 4 to 15 clocks with minimized execution timing (<u>EU1 only</u>)”. Section 1.3.5, “Memory Management Unit (MMU) Features”, deleted “Byte ordering (endianness) configurable on a per-page basis” Section 1.3.6, “System Bus (Core Complex Interface) Features” - Changed “32-bit address bus plus attributes and control on each bus” to “32-bit address bus, <u>64-bit data bus</u>, plus attributes and control on each bus”. - Added “Support for HCLK running at a slower rate than CPU clock”.

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 2, "Register Model"	Chapter 2, "Register Model" - In Figure 2-1, "e200z760 Supervisor Mode Programmer's Model", corrected misaligned register names for Memory Management Registers, Control & Configuration. Section 2.4.1, "Machine State Register (MSR)" - In Figure 2-5, "Machine State Register (MSR)", corrected bit 28 to "—" and bit 29 to "PMM".
Chapter 3, "Instruction Model"	Section 3.13, "Enhanced Reservations" Throughout, corrected code listings to display arrow symbols, subscripts, and superscripts properly, such as "if X-mode then EA = 320 (a + GPR(RB))32:63" to "if X-mode then EA = 320 (a + GPR(RB))32:63".
Chapter 4, "Instruction Pipeline and Execution Timing"	Section 4.1, "Overview of Operation" - In Figure 4-1, "e200z7 Block Diagram", corrected "Single-Instruction, In-Order Dispatch" to "Dual-Instruction, In-Order Dispatch" and "Single-Instruction, In-Order Write Back" to "Dual-Instruction, In-Order Write Back" Section 4.3.2, "Instruction Prefetch Buffers and Branch Target Buffer" - Changed "Certain other branches do not allocate BTB entries: <u>blr</u>, <u>bclr</u>, <u>bctr</u>, <u>bcctr</u>." to "Certain other branches do not allocate BTB entries: <u>bctr</u>, <u>bcctr</u>."
Chapter 5, "Embedded Floating-Point Unit"	Section 5.3.1.1, "Single-Precision Floating-point Format" - Added missing title to Figure 5-2, "Single-Precision Data Format". Section 5.3.4, "Embedded Scalar Single-Precision Floating-Point Instructions" - For the <code>evfscfh</code> instruction, corrected code listing as follows: - Changed line 15 "resulth = fhsign 151 // like-signed zero value" to "resulth = fhsign 0b11110 100 // max value". - Changed line 18 "resulth = fhsign 0b00001 100 // min value" to "resulth = fhsign 150 // like-signed zero value". Section 5.6, "Instruction Forms and Opcodes" In Table 5-16, "Embedded Vector Floating-Point Instruction Opcodes" and Table 5-17, "Embedded Scalar Single-Precision Floating-Point Instruction Opcodes" for the lines whose Instruction value is "—", changed the value of Opcode Bits, 0-5 to "4"
Chapter 6, "Signal Processing Extension (SPE)"	Section 6.1, "Nomenclature and Conventions" Added "Due to historical precedent, the terms SPE and SIMD are sometimes used interchangeably."
Chapter 7, "Interrupts and Exceptions"	Section 7.5, "Interrupt Vector Offset Registers (IVORxx)" - In Figure 7-5, "e200 Interrupt Vector Offset Register (IVOR)", added missing "—" for bits 28-31. Section 7.6, "Interrupt Definitions" - In Table 7-8, "Interrupts", changed IVOR Number "IVOR17", "IVOR18", "IVOR19", and "IVOR20" to "IVOR32", "IVOR33", "IVOR34", and "IVOR35", respectively. Section 7.6.17, "System Reset Interrupt" - In Table 7-30, "System Reset Interrupt—Register Settings" - Added row for Register "CSRR0" with Setting Description "Undefined". - Changed Vector notation "[p_rstbase[0:29]] 2'b00" to "[p_rstbase[0:29]] 0b00" for consistency. Section 7.7.1, "Exception Priorities" - In Table 7-35, "Zen Exception Priorities", for entries whose Exception describes the Debug procedure, added numbers for Cause entries.

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 8, "Performance Monitor"	Section 8.3.7, "Local Control B Registers (PMLCb0–PMLCb3)" • In Table 8-6, "PMLCb0–PMLCb3 Field Descriptions", for Bits "9:12" and "14:17", in the Description field, changed "<u>1100</u>Trigger-on based on watchpoint #28 occurrence", "<u>1101</u>Trigger-on based on watchpoint #29 occurrence", and "<u>1110</u>-1111 Reserved" to "<u>1000</u>Trigger-on based on watchpoint #24 occurrence", "<u>1001</u>Trigger-on based on watchpoint #25 occurrence", and "<u>1100</u>-1111 Reserved", respectively. Section 8.7, "Event Selection" • In Table 8-10, "Performance Monitor Event Selection", for items Number "Com:146", "Com:147", and "Com:148", changed all values to "—".

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 9, "L1 Cache"	Section 9.3, "Cache Lookup" - Changed "Subsequent double-words may be streamed to the CPU if they have been requested and streaming is enabled via the L1CSR0 register, ..." to "Subsequent double-words may be streamed to the CPU if they have been requested, ...". Section 9.4.2, "L1 Cache Control and Status Register 1 (L1CSR1)" - Added "The SPR number for L1CSR1 is 1011 in decimal." - In Table 9-2, "L1CSR1 Field Descriptions", deleted Bits "14" row, ISTRM. Section 9.4.4, "L1 Cache Configuration Register 1 (L1CFG1)" - Changed "e200z7" to "e200z760n3". - In Figure 9-7, "L1 Cache Configuration Register 1 (L1CFG1)", and changed "Access: Read/Write" to "Access: Read only". Section 9.7.4, "Cache Miss Access Ordering" - Changed "e200z7" to "e200z760n3". Section 9.7.9.2, "L1 Flush/Invalidate Register 1 (L1FINV1)" - Corrected title of Figure 9-9, "L1 Flush/Invalidate Register 1 (L1FINV1)". Section 9.9, "Push and Store Buffers" - Deleted "For the AXI interface version, the burst write bus transaction to write the contents of the push buffer into memory is performed in parallel with miss linefill operations, without waiting for the linefill to complete." Section 9.12, "Cache Line Locking/Unlocking Unit" - In Table 9-9, "Cache Line Locking/Unlocking Unit Instructions", corrected Acronym "dcbtstls" to "dcbtstls". Section 9.12.2, "Instruction Details" - In Table 9-10, "Cache Line Locking/Unlocking Unit Instructions", corrected Acronym "dcbtstls" to "dcbtstls". Section 9.13.1, "Exception Conditions for Cache Instructions" - Added the following Notes to Table 9-11, "Special Case Handling": "Notes: - Priority decreases from left to right - Cache operations that do not set or clear locks ignore the value of the CT field - "dash" indicates executes normally - "NOP" indicates treated as a no-op - DSI = data storage interrupt; ALI = alignment interrupt; DTLB = data TLB interrupt - DCUL, ICUL = no-op, and set L1CSR0[CUL] - DCLO, ICLO = no-op, and set L1CSR0[CLO] - DLK, ILK = data storage interrupt (DSI) and set ESR[DLK] or ESR[ILK] - MC = Machine Check and update MCAR" Section 9.13.2, "Transfer Type Encodings for Cache Management Instructions" - In Table 9-12, "Transfer Type Encoding", added "1" to table header "p_d_ttype[0:5]" and added footnote "1 p_ttype[5] 'e' is set to set to 0." Section 9.19.3, "Cache Debug Access Control Register (CDACNTL)" - In Figure 9-10, "Cache Debug Access Control Register (CDACNTL)", and added DCR number "DCR 351". Section 9.19.3.1, "Cache Debug Access Data Register (CDADATA)" - In Figure 9-11, "Cache Debug Access Data Register (CDADATA)", and added DCR number "DCR 350". Added new Section 9.20, "Hardware Debug (Cache) Control Register 0".

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 10, "Memory Management Unit"	Section 10.4.1, "MMU Configuration Register (MMUCFG)" • In Figure 10-4, "MMU Configuration Register (MMUCFG)", deleted Reset value "All zeros". Section 10.4.3, "TLB1 Configuration Register (TLB1CFG)" • In Figure 10-6, "TLB1 Configuration Register (TLB1CFG)", deleted Reset value "All zeros". Section 10.7.3, "MMU Assist Registers (MAS)" • In Figure 10-10, "MMU Assist Register 1 (MAS1)", added missing field name for bits 20-24 "TSIZE". Section 10.9.1, "Transfer Type Encodings for MMU Control Instructions" • In Table 10-16, "Transfer Type Encoding", added "1" to table header "p_d_ttype[0:5]" and added footnote "1 p_ttype[5] 'e' is set to set to 0." Section 10.11, "External Translation Alterations for Realtime Systems" • Corrected "Those entries within entries 0–15 programmed with a TID value of 0b1111nm11..." to "Those entries within entries 0–15 programmed with a TID value of 0b1111nm11..."

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 11, “External Core Complex Interfaces”	Section 11.1, “Signal Index” - In Table 11-1, “Interface Signal Definitions”, added section “External Translation Alteration Signals” with two entries, Signal Name “p_extpid_en” and “p_extpid[6:7]”. - For Signal Name “p_stop”, added “—” for Reset Value. - For “JTAG-Related Signals” section, added entry Signal Name “j_key_in”. Section 11.2.13.1, “Cache Tag Error Out (p_[d,i]_cache_tagerr_out)” - Added “When L1CSR0[DCEA]/L1CSR1[ICEA] indicates machine check generation on error, assertion of this signal indicates a machine check will be signaled for the access, or for dcbi/icbi operations, indicates that a remote invalidation of one or more cache lines should occur. When L1CSR0[DCEA]/L1CSR1[ICEA] indicates auto-invalidation on error, assertion of this signal indicates that the cache will insert an additional cycle to perform auto-invalidation on cache ways with uncorrectable tag errors, and to correct tags in ways with correctable errors.” Added new section Section 11.2.14, “External Translation Alteration Signals” with subsections Section 11.2.14.1, “External PID Enable (p_extpid_en)” and Section 11.2.14.2, “External PID In (p_extpid[6:7])”. Section 11.2.22.6, “Watchpoint Events (jd_watchpt[0:26])” - Corrected title from “jd_watchpt[0:29]” to “jd_watchpt[0:26]”. - Deleted sentence “DEVNT-, DTC-based, and performance monitor watchpoints are also supported.” Section 11.2.24, “Development Support (Nexus 3) Signals” - In Table 11-21, “e200 Development Support (Nexus) Signals”, changed Signal “nex_wevto[3:0]” to “nex_wevto[2:0]”. Section 11.2.25.5, “JTAG/OnCE Test Reset (j_trst_b)” - In Table 11-23, “JTAG Signals Used to Support External Registers”, added entry Signal Name “j_key_in”. Added new Section 11.2.25.12, “Key Data In (j_key_in)”. Section 11.3.2.1, “Basic Read Transfer Cycles” - In Figure 11-7, “Basic Read Transfers”, added caption “Single cycle reads, full pipelining”. Section 11.3.2.7, “Burst Accesses” - In Figure 11-25, “Burst Read Error Termination, Burst Write Substituted”, added caption “Burst Read with error termination, Burst write”. Section 11.3.3, “Memory Synchronization Control Operation”, inserted missing figures Figure 11-28, “Memory Sync Operation (snoop queue empty)” and Figure 11-29, “Memory Sync Operation (2nd msync back-to-back)”. Section 11.3.5.1, “Stop Mode Entry/Exit and Snoop Ready Signaling” - In Figure 11-54, “Stop Mode Exit, p_snp_rdy Operation”, added caption “Snoop interface operation - stop mode exit operation”. Section 11.3.6.1, “Debug Entry Cross-signaling” - In Figure 11-55, “Debug Entry Cross-Signaling Interface, non-Lockstep Mode”, added caption “Debug exit, non-Lockstep operation, CPU 0 sees OCMD “go” first, debug mode exit not synchronized”.

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 13, "Debug Support"	Section 13.3.3.1, "Debug Control Register 0 (DBCR0)" - Added to footnote to Figure 13-4, "DBCR0 Register" "If DBCR0[EDM]=1, DBERC0 masks off hardware-owned resources (other than RST) from reset by p_reset_b, and only software-owned resources indicated by DBERC0 and the DBCR0[RST] field will be reset by p_reset_b." - In Table 13-6, "DBCR0 Bit Definitions" - For the Description of Bit "0", changed "{DBCR0-6, DBCNT, IAC1-8, DAC1-2}" to "{DBCR0, DBCNT, IAC1, DAC1-2}" . - For the Description of Bit "1", changed "Debug events do not affect DBSR unless EDM is set." to "Debug events do not affect DBSR". Section 13.3.3.8, "Debug Status Register (DBSR)" - In Figure 13-11, "DBSR Register" changed bit field name "IAC4" to "IAC4-8". Section 13.4, "External Debug Support" - In NOTE, changed "EDBCR0[EDM]" to "EDBCR0[EDM]/DBCR0[EDM]". Section 13.4.5.5, "Watchpoint Events (jd_watchpt[0:29])" - Changed "Watchpoint events are conditioned by the settings in the DBCR0, DBCR1, and DBCR2 registers, as well as by the DEVENT register, <u>the DTC/DTSA/DTEA registers</u>, and the Performance Monitor control register settings" to "Watchpoint events are conditioned by the settings in the DBCR0, DBCR1, and DBCR2 registers, as well as by the DEVENT register, and the Performance Monitor control register settings". Section 13.4.6.2, "e200 OnCE Command Register (OCMD)" - Changed "EDBCR0[EDM]" to "EDBCR0[EDM]/DBCR0[EDM]". Section 13.5, "Watchpoint Support" - Changed "Certain watchpoints (DEVNT-based <u>and DTC-based</u>)..." to "Certain watchpoints (DEVNT-based)..." Changed "The DEVNT-based <u>and DTC-based</u> watchpoints are..." to "The DEVNT-based watchpoints are..."
Chapter 14, "Nexus 3 Module"	Chapter 14, "Nexus 3 Module" - Throughout chapter, changed "the IEEE-ISTO 5001-2008 standard" to "the IEEE-ISTO 5001 standard" except for certain occurrences. Section 14.1.1, "Terms and Definitions" - In Table 14-1, "Terms and Definitions" changed Term "Nexus3" to "Nexus1". Section 14.1.2, "Feature List" - Changed "<u>Four (4)</u> additional Watchpoint Event output pins (nex_wevto[3:0]) for SoC use" to "<u>Three (3)</u> additional Watchpoint Event output pins (nex_wevto[2:0]) for SoC use" Section 14.4.4, "Nexus Development Control Register 2 (DC2)" - In Figure 14-5, "Development Control Register 2 (DC2)" changed "WEVTO[3]C" to "—". - In Table 14-12, "Development Control Register 2 Fields" changed Bits "DC2[31-24]" Name "EWC" to Bits "31-28" Name "—" and Description to "Reserved". Section 14.4.5, "Nexus Development Control Register 3 (DC3)" - In Figure 14-6, "Development Control Register 3 (DC3)" changed "WEVTO[3]C" to "—". - In Table 14-13, "Development Control Register 3 Fields" changed Bits "31-28" from Name "WEVTO[3]C" to "—" and Description to "Reserved".

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 14, "Nexus 3 Module" (continued)	Section 14.4.8, "Watchpoint Trigger Registers (WT, PTSTC, PTETC, DTSTC, DTETC)" - In Figure 14-10, "Program Trace Start Trigger Control (PTSTC) Register", Figure 14-11, "Program Trace End Trigger Control (PTETC) Register", Figure 14-12, "Data Trace Start Trigger Control (DTSTC) Register", and Figure 14-13, "Data Trace End Trigger Control (DTETC) Register", moved start of PTST, PTET, DTST, and DTET bit field, respectively, from bit 29 to 26. - In Table 14-17, "Program Trace Start Trigger Control Register Fields", Table 14-18, "Program Trace End Trigger Control Register Fields", Table 14-19, "Data Trace Start Trigger Control Register Fields", and Table 14-20, "Data Trace End Trigger Control Register Fields", changed start of PTST, PTET, DTST, and DTET bit field, respectively, from bit 29 to 26. Changed Description to represent 27 bits. Section 14.4.9, "Nexus Watchpoint Mask Register (WMSK)" - In Figure 14-14, "Watchpoint Mask Register", moved start of WEM bit field from bit 29 to 26. - In Table 14-21, "Watchpoint Mask Register Fields", changed start of WEM bit field from bit 29 to 26. Changed Description to represent 27 bits. Section 14.4.14, "Read/Write Access Control/Status (RWCS)" - In Table 14-25, "Read/Write Access Control/Status Register Fields", corrected Bits "RWCS[21-16]" to "RWCS[17-16]". Section 14.7.4, "Nexus Message Priority" - In Table 14-29, "Message Type Priority and Message Dropped Responses", added footnote for Message Dropped Response of 5th entry: "Message will always be dropped if program trace is enabled, and program correlation messages for PID0 /mtmsr IS messages are not masked (Event Code = 0101). No error message is sent for this case since the PID value is contained in the higher priority message." Section 14.11.3.10, "Program Trace Synchronization Messages" - In Table 14-34, "Program Trace Exception Summary", for Exception Condition "Collision Priority" changed the Instruction 0 priority description from "WPM → PCM[PIDMSG] → DQM ..." to "WPM → DQM → PCM[PIDMSG]...". Section 14.11.5, "Program Trace Timing Diagrams (2 MDO/1 MSEO Configuration)" - In Figure 14-33, "Program Trace—Indirect Branch Message (Traditional)", for MDO[1:0] changed the 6th value from "00" to "10". - In Figure 14-34, "Program Trace—Indirect Branch Message (History)", for MDO[1:0] changed the 6th value from "00" to "01". Section 14.12.4, "Data Trace Timing Diagrams (8 MDO/2 MSEO Configuration)" - In Figure 14-40, "Data Trace—Data Write Message", for MDO[7:0] changed the 2nd value from "01001000" to "10010100" and changed the 3rd value from "000101001" to "01010010". - In Figure 14-41, "Data Trace—Data Read with Sync Message", deleted "addr type = 0, " Section 14.13.3, "Data Acquisition Trace Event" - Added Figure 14-42, "Data Acquisition Message Format". Section 14.14, "Watchpoint Trace Messaging" - In Figure 14-43, "Watchpoint Message Format", changed "(1–30 bits)" to "(1–27 bits)" and "Variable length = 11–40 bits" to "Variable length = 11–37 bits". In Table 14-37, "Watchpoint Source Encoding", changed column header "Watchpoint Source (1-30 bits)" to "Watchpoint Source (1-27 bits)". Changed Description to represent 27 bits.

Table B-2. Changes between revisions 0 and 1 (continued)

Chapter	Description
Chapter 14, "Nexus 3 Module" (continued)	Section 14.14.1, "Watchpoint Timing Diagram (2 MDO/1 MSEO configuration)" - Changed title of Figure 14-44, "Watchpoint Message and Watchpoint Error Message" to "Watchpoint Message and Watchpoint Error Message". Section 14.15, "Nexus 3 Read/Write Access to Memory-Mapped Resources" - Added paragraph "Nexus 3 read/write accesses are run as privileged data non-cacheable, non-global accesses by default, and drive the p_d_hprot[5:0] bus access attributes to 0b000011 and the p_d_gbl access attribute to 0 accordingly. The RWCS[ATTR] field is provided to allow a portion of these default values to be modified when performing read or write accesses using the Nexus 3 Read/Write access mechanism." Section 14.16.1, "Pins Implemented" - Changed "(4) nex_wevto[3:0] pins" to "(3) nex_wevto[2:0] pins" - In Table 14-39, "Nexus 3 Auxiliary Pins", changed "nex_wevto[3:0]" to "nex_wevto[2:0]" and "Watchpoint Event Out 3-0" to "Watchpoint Event Out 2-0".
Section Appendix A, "Register Summary"	Section Appendix A, "Register Summary" - In Figure A-1, "e200z760 Supervisor Mode Programmer's Model", corrected misaligned register names for Memory Management Registers, Control & Configuration. - In Figure A-5, "Machine State Register (MSR)", corrected bit 28 to "—" and bit 29 to "PMM". - In Figure A-18, "e200 Interrupt Vector Offset Register (IVOR)", added missing "—" for bits 28-31. - In Figure A-26, "L1 Cache Configuration Register 1 (L1CFG1)", changed "Access: Read/Write" to "Access: Read only". - Corrected title of Figure A-28, "L1 Flush/Invalidate Register 1 (L1FINV1)". - In Figure A-29, "MMU Configuration Register (MMUCFG)", deleted Reset value "All zeros". - In Figure A-31, "TLB1 Configuration Register (TLB1CFG)", deleted Reset value "All zeros". - In Figure A-35, "MMU Assist Register 1 (MAS1)", added missing field name for bits 20-24 "TSIZE". - Added to footnote to Figure A-42, "DBCR0 Register" "If DBCR0[EDM]=1, DBERC0 masks off hardware-owned resources (other than RST) from reset by p_reset_b, and only software-owned resources indicated by DBERC0 and the DBCR0[RST] field will be reset by p_reset_b." - In Figure A-49, "DBSR Register", changed bit field name "IAC4" to "IAC4-8".
Section Appendix B, "Revision History"	Added Section B.1, "Changes between revisions 0 and 1"